

A NOTATIONS

Table 4: Notations.

SYMBOL	DEFINITION
s_t	THE ENVIRONMENT STATE AT TIMESTEP t
a_t	THE ACTION INTERACTING WITH THE ENVIRONMENT AT TIMESTEP t
r_t	THE REWARD FEEDBACK FROM THE ENVIRONMENT AT TIMESTEP t
$\tilde{s}_t$	THE STATE OBSERVED BY THE AGENT AT TIMESTEP t
$\tilde{a}_t$	THE ACTION TAKEN BY THE AGENT AT TIMESTEP t
$\tilde{r}_t$	THE REWARD RECEIVED BY THE AGENT AT TIMESTEP t
p^a	ACTION DELAY DISTRIBUTION
p^o	OBSERVATION DELAY DISTRIBUTION
p^{joint}	THE DISTRIBUTION OF THE SUM OF ACTION DELAY AND OBSERVATION DELAY
δ_a	MAXIMUM VALUE OF THE ACTION DELAY DISTRIBUTION
δ_o	MAXIMUM VALUE OF THE OBSERVATION DELAY DISTRIBUTION
δ^{joint}	MAXIMUM VALUE OF p^{joint}
d_p	WASSERSTEIN METRIC
$\tilde{d}_p$	THE MAXIMAL FORM OF THE WASSERSTEIN METRIC

B PROOF

B.1 Proposition 1 Derivation

Proof. Under pure action delay, the stochastic delay representation $Z(\tilde{s}_t, \tilde{a}_t)$ at timestep t is expressed as:

$$Z(\tilde{s}_t, \tilde{a}_t) = \sum_{i=1}^{\delta_a} p_i^a \cdot (\tilde{r}_{t+i} + \gamma Z(\tilde{s}_{t+i+1}, \tilde{a}_{t+i+1})). \quad (9)$$

Under pure observation delay, it takes the form:

$$Z(\tilde{s}_t, \tilde{a}_t) = \sum_{j=1}^{\delta_o} p_j^o \cdot (\tilde{r}_{t+j} + \gamma Z(\tilde{s}_{t+j+1}, \tilde{a}_{t+j+1})). \quad (10)$$

In the presence of both delays, assuming independence of i and j as $\mathbb{P}(i, j) = p_i^a \cdot p_j^o$, the combined version of stochastic delay representation becomes:

$$Z(\tilde{s}_t, \tilde{a}_t) = \sum_{i=1}^{\delta_a} \sum_{j=1}^{\delta_o} p_i^a \cdot p_j^o \cdot (\tilde{r}_{t+(i+j)} + \gamma Z(\tilde{s}_{t+(i+j)+1}, \tilde{a}_{t+(i+j)+1})). \quad (11)$$

Let $k = i + j$ and define the joint delay distribution as:

$$p_k^{joint} = \sum_{\substack{i+j=k \\ i \in [1, \delta_a], j \in [1, \delta_o]}} p_i^a \cdot p_j^o, \quad (12)$$

with the total maximum delay denoted as $\delta_{joint} = \delta_a + \delta_o$. To simplify notation and unify bounds across formulations, we allow k to range from 1 to δ_{joint} and define $p_1^{joint} = 0$ to cover the case where no such (i, j) exists with $i + j = 1$. This does not affect the value of the summation but allows consistent indexing.

Substituting into Eq. (11) yields the unified form:

$$Z(\tilde{s}_t, \tilde{a}_t) = \sum_{k=1}^{\delta_{joint}} p_k^{joint} \cdot (\tilde{r}_{t+k} + \gamma Z(\tilde{s}_{t+k+1}, \tilde{a}_{t+k+1})). \quad (13)$$

Comparing Eq. (13) with Eqs. (9) and (10), we conclude that $Z(\tilde{s}_t, \tilde{a}_t)$ depends only on the aggregated delay k and its probability p_k^{joint} , and not on whether the delay originates from action or observation. Therefore, $Z(\tilde{s}_t, \tilde{a}_t)$ is invariant under the decomposition of k into i and j , establishing the equivalence of delay types in the stochastic delay representation.

B.2 Theorem 1 Derivation

The proof of Theorem 1 follows the classic analytical framework of Policy Iteration, with its structure sequentially built upon three lemmas. First, we lay the theoretical cornerstone with a core equivalence, Lemma 1, which demonstrates that our stochastic delay representation operator is equivalent in expectation to the marginalized operator of a theoretically sound State-Augmented MDP, thus establishing the validity of our method within a Markovian framework. Building upon this foundation, Lemma 2 proves the convergence of the policy evaluation process, while Lemma 3 guarantees the monotonicity of the policy improvement step. Finally, by combining the convergent policy evaluation with the monotonic policy improvement, we prove that the overall policy iteration process converges to the optimal policy, thus completing the proof of Theorem 1.

The proof of Theorem 1 requires justifying the use of Bellman-style operators on the non-Markovian observation sequence. Our proof accomplishes this by establishing a fundamental operator-level equivalence. We formally demonstrate that the Stochastic Delay Representation evaluation operator is equivalent in expectation to the marginalized Bellman operator from a theoretically sound State-Augmented MDP. This equivalence serves as the formal bridge that allows the powerful tools of Bellman analysis, such as contraction and policy improvement, to be validly applied to our method.

1. Analysis Framework and Assumptions.

- **Delay Type:** We focus on environments with observation delay only in Lemma 1. As established in Proposition 1, this does not compromise the generality of our main theorem due to the equivalence of delay types.
- **Policy Class:** We restrict attention to observation-based policies ($\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$), which aligns with the representational capacity of our stochastic delay representation framework. While augmented-MDP policies could in principle exploit richer history information, our result shows that within the considered class, our method admits a sound Bellman-style operator characterization.
- **Formal Assumptions:** Our analysis relies on the following: (i) The effective delay range is bounded by $\delta_o < \infty$. (ii) The observation delay distribution p^o is known, stationary, and independent of the state-action pair (s, a) . (iii) The action space $|\mathcal{A}|$ is finite and the rewards are bounded.

2. Relevant Process Definitions.

- Observation-Delay POMDP: The agent operates in a Partially Observable MDP where its observation $\tilde{s}_t$ is a delayed past state from the true, unobserved MDP $\mathcal{M}$, while its action $\tilde{a}_t$ applies to the current true state s_t .
- Augmented MDP: To recover the Markov property, we follow [30] and define the augmented state as $\tilde{s}_t = (\tilde{s}_t, \tilde{H}_t^{act})$, where $\tilde{H}_t^{act} = (\tilde{a}_{t-1}, \dots, \tilde{a}_{t-\delta_o})$ is the recent action-history. This construction yields a valid MDP $\mathcal{M}' = (\tilde{\mathcal{S}}, \mathcal{A}, P', R', \gamma)$ with a well-defined Bellman operator $\mathcal{T}_{\mathcal{M}'}$.

To bridge the gap between the value function on the augmented space $Q_{\mathcal{M}'}$, and the one on the observable space, we define an aggregated value function, Q_{agg} . It serves as the theoretical target for our method, representing the best possible value estimate when conditioned only on the delayed observation.

Lemma 1 (Marginalized Equality). *Let $Q_{\mathcal{M}'}^\pi : \tilde{\mathcal{S}} \times \mathcal{A} \rightarrow \mathbb{R}$ be the action-value function for a policy π from the restricted class. We define the aggregated value function Q_{agg}^π as:*

$$Q_{agg}^\pi(\tilde{s}, \tilde{a}) := \mathbb{E}_{\tilde{H}^{act} \sim \mu(\cdot|\tilde{s})} [Q_{\mathcal{M}'}^\pi((\tilde{s}, \tilde{H}^{act}), \tilde{a})] \quad (14)$$

Here, $\mu(\cdot|\tilde{s})$ is the conditional distribution of $\tilde{H}_t^{act}$ given $\tilde{s}_t$. This conditional distribution is well-defined because $(\tilde{s}_t, \tilde{H}_t^{act})$ are jointly distributed under the stationary process induced by (π, p^o) , hence conditioning on $\tilde{s}_t$ is valid probabilistic inference. For any fixed policy π from the aforementioned class, the stochastic delay representation (SDR) evaluation operator $\mathcal{T}_{SDR}^\pi$ and the augmented Bellman evaluation operator $\mathcal{T}_{\mathcal{M}'}^\pi$ satisfy the following exact identity:

$$(\mathcal{T}_{SDR}^\pi Q_{agg}^\pi)(\tilde{s}_t, \tilde{a}_t) = \mathbb{E}_{\tilde{H}_t^{act} \sim \mu(\cdot|\tilde{s}_t)} \left[(\mathcal{T}_{\mathcal{M}'}^\pi Q_{\mathcal{M}'}^\pi)((\tilde{s}_t, \tilde{H}_t^{act}), \tilde{a}_t) \right] \quad (15)$$

Proof. We prove the identity by expanding its right-hand side and showing that it equals the left-hand side.

The left-hand side of the identity is, by definition, the expected-value stochastic delay representation operator applied to Q_{agg}^π :

$$(\mathcal{T}_{SDR}^\pi Q_{agg}^\pi)(\tilde{s}_t, \tilde{a}_t) = \mathbb{E} [r_t + \gamma \mathbb{E}_{\tilde{a}_{t+1} \sim \pi(\cdot|\tilde{s}_{t+1})} [Q_{agg}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] \mid \tilde{s}_t, \tilde{a}_t] \quad (16)$$

where $r_t = R(s_t, \tilde{a}_t)$ denotes the instantaneous reward generated in the environment at timestep t (which the agent receives later as $\tilde{r}_{t+k}$).

Now, we expand the right-hand side of the identity. By the definition of the augmented operator $\mathcal{T}_{\mathcal{M}'}^\pi$, and our assumption that the policy π depends only on the delayed observation component, this term is:

$$\mathbb{E}_{\tilde{H}_t^{act} \sim \mu(\cdot|\tilde{s}_t)} \left[\mathbb{E} [r_t + \gamma \mathbb{E}_{\tilde{a}_{t+1} \sim \pi(\cdot|\tilde{s}_{t+1})} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] \mid \tilde{s}_t = (\tilde{s}_t, \tilde{H}_t^{act}), \tilde{a}_t] \right] \quad (17)$$

By the Law of Total Expectation, averaging the inner conditional expectation over $\tilde{H}_t^{act}$ marginalizes out the history component from the condition, yielding:

$$\mathbb{E} [r_t + \gamma \mathbb{E}_{\tilde{a}_{t+1} \sim \pi(\cdot|\tilde{s}_{t+1})} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] \mid \tilde{s}_t, \tilde{a}_t] \quad (18)$$

Then, we focus on the future value term, which is the total expectation over the randomness of $\tilde{s}_{t+1}$ and the policy's selection of $\tilde{a}_{t+1}$:

$$\mathbb{E}_{\tilde{a}_{t+1} \sim \pi(\cdot|\tilde{s}_{t+1})} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] \quad (19)$$

We can decompose this total expectation by first conditioning on a specific outcome of the random variables $(\tilde{s}_{t+1}, \tilde{a}_{t+1})$ using the Law of Total Expectation:

$$\mathbb{E} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] = \mathbb{E}_{\tilde{s}_{t+1}, \tilde{a}_{t+1}} \left[\mathbb{E} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1}) \mid \tilde{s}_{t+1}, \tilde{a}_{t+1}] \right] \quad (20)$$

The inner expectation now averages over the only remaining randomness in $\tilde{s}_{t+1} = (\tilde{s}_{t+1}, H_{t+1}^{act})$, which is the history component H_{t+1}^{act} . By the definition of Q_{agg}^π , this inner expectation is precisely $Q_{agg}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})$:

$$\mathbb{E} [Q_{\mathcal{M}'}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1}) \mid \tilde{s}_{t+1}, \tilde{a}_{t+1}] = \mathbb{E}_{H_{t+1}^{act} \sim \mu(\cdot|\tilde{s}_{t+1})} [Q_{\mathcal{M}'}^\pi((\tilde{s}_{t+1}, H_{t+1}^{act}), \tilde{a}_{t+1})] = Q_{agg}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1}) \quad (21)$$

Now, we substitute this result back into the full expression for the right-hand side. This substitution resolves the future value term, yielding:

$$\mathbb{E} [r_t + \gamma \mathbb{E}_{\tilde{a}_{t+1} \sim \pi(\cdot|\tilde{s}_{t+1})} [Q_{agg}^\pi(\tilde{s}_{t+1}, \tilde{a}_{t+1})] \mid \tilde{s}_t, \tilde{a}_t] \quad (22)$$

This final expression is now identical to the definition of the left-hand side of the identity, as given in Eq. (14).

Therefore, we have rigorously shown that the right-hand side of the identity equals the left-hand side. The identity holds. By induction on the number of Bellman backups, this one-step identity extends to multi-step backups. As the number of iterations approaches infinity, and since the contraction property (Lemma 2) guarantees a fixed point for both operators, this equivalence holds for the evaluation fixed points. Thus, stochastic delay representation evaluation converges to the marginalized fixed point of $\mathcal{M}'$. Note that this fixed point corresponds to the best achievable value function within the restricted policy class, which matches the representational scope of stochastic delay representation.

Then, we provide the remaining proof of Theorem 1, which characterizes the properties of the stochastic delay representation method, building upon existing theoretical results in distributional reinforcement learning [3, 9, 24].

The stochastic delay representation with distributional returns operator derived from the Eq. (4) can be defined as

$$\mathcal{T}^\pi Z(\tilde{s}, \tilde{a}) = \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \left(\tilde{R}_{i+j}(\tilde{s}_{i+j}) + \gamma P^\pi Z(\tilde{s}_{i+j}, \tilde{a}_{i+j}) \right) \quad (23)$$

where $P^\pi : \mathcal{Z} \rightarrow \mathcal{Z}$ is a state transition operator under policy π , with $\mathcal{Z}$ defined as the distributional value function space, and $P^\pi Z(\tilde{s}, \tilde{a}) \stackrel{D}{=} Z(\tilde{s}', \tilde{a}')$, where

$$\tilde{s}' \sim \mathbb{E}_{\substack{i \sim p_i^a \\ j \sim p_j^o}} P(\cdot \mid s_{-j}, \tilde{a}_{-i-j}) \quad (24)$$

and $\tilde{a}' \sim \pi(\cdot \mid \tilde{s}')$. The subscripts indicate the temporal relationship of states and actions relative to the current timestep. Specifically, $(\tilde{s}_{i+j}, \tilde{a}_{i+j})$ represents the state-action pair $i+j$ timesteps after the current pair $(\tilde{s}, \tilde{a})$, s_{-j} denotes the environment state i timesteps prior to the current pair, and $\tilde{a}_{-i-j}$ refers to the action taken $i+j$ timesteps prior to the current pair. We view the reward function as a random vector $\tilde{R}_{i+j} \in \mathcal{Z}$.

Lemma 2: (Stochastic Delay Representation with Distributional Returns Policy Evaluation): Consider the stochastic delay representation with distributional returns backup operator $\mathcal{T}^\pi$, and a state-action distribution function $\mathcal{Z}_0(Z_0(\tilde{s}, \tilde{a}) \mid \tilde{s}, \tilde{a}) : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(Z_0(\tilde{s}, \tilde{a}))$, which maps a state-action pair $(\tilde{s}, \tilde{a})$ to a distribution over random state-action returns $Z_0(\tilde{s}, \tilde{a})$. Define $Z_{i+1}(\tilde{s}, \tilde{a}) = \mathcal{T}^\pi Z_i(\tilde{s}, \tilde{a})$, where $Z_{i+1}(\tilde{s}, \tilde{a}) \sim \mathcal{Z}_{i+1}(\cdot \mid \tilde{s}, \tilde{a})$. Then, the sequence $\mathcal{Z}_i$ will converge to $\mathcal{Z}^\pi$ as $i \rightarrow \infty$.

Proof. As proved in [3], $\bar{d}_p$ is a metric over value distributions, which can be formulated as

$$\bar{d}_p(Z_1, Z_2) = \sup_{s,a} d_p(Z_1(s, a), Z_2(s, a)) \quad (25)$$

where d_p is the Wasserstein metric, which has the following properties:

$$d_p(aU, aV) \leq |a|d_p(U, V) \quad (P1)$$

$$d_p(A + U, A + V) \leq d_p(U, V) \quad (P2)$$

$$d_p\left(\sum_{i=1}^n U_i, \sum_{i=1}^n V_i\right) \leq \sum_{i=1}^n d_p(U_i, V_i) \quad (P3)$$

where a is a scalar, U and V are random variables, while U_i and V_i are independent.

Consider $Z_1, Z_2 \in \mathcal{Z}$. By definition,

$$\bar{d}_p(\mathcal{T}^\pi Z_1, \mathcal{T}^\pi Z_2) = \sup_{\tilde{s}, \tilde{a}} d_p(\mathcal{T}^\pi Z_1(\tilde{s}, \tilde{a}), \mathcal{T}^\pi Z_2(\tilde{s}, \tilde{a})). \quad (26)$$

By the properties of d_p , we can obtain that

$$\begin{aligned} & d_p(\mathcal{T}^\pi Z_1(\tilde{s}, \tilde{a}), \mathcal{T}^\pi Z_2(\tilde{s}, \tilde{a})) \\ &= d_p\left(\sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \left(\tilde{R}_{i+j}(\tilde{s}_{i+j}) + \gamma P^\pi Z_1(\tilde{s}_{i+j}, \tilde{a}_{i+j})\right), \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \left(\tilde{R}_{i+j}(\tilde{s}_{i+j}) + \gamma P^\pi Z_2(\tilde{s}_{i+j}, \tilde{a}_{i+j})\right)\right) \\ &\leq d_p\left(\sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma P^\pi Z_1(\tilde{s}_{i+j}, \tilde{a}_{i+j}), \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma P^\pi Z_2(\tilde{s}_{i+j}, \tilde{a}_{i+j})\right) \\ &\leq \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma d_p(P^\pi Z_1(\tilde{s}_{i+j}, \tilde{a}_{i+j}), P^\pi Z_2(\tilde{s}_{i+j}, \tilde{a}_{i+j})) \\ &\leq \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma \sup_{\tilde{s}'_{i+j}, \tilde{a}'_{i+j}} d_p(Z_1(\tilde{s}'_{i+j}, \tilde{a}'_{i+j}), Z_2(\tilde{s}'_{i+j}, \tilde{a}'_{i+j})) \end{aligned} \quad (27)$$

By utilizing the definition of Eq. (26) and the results of Eq. (27), we obtain

$$\begin{aligned} \bar{d}_p(\mathcal{T}^\pi Z_1, \mathcal{T}^\pi Z_2) &= \sup_{\tilde{s}, \tilde{a}} d_p(\mathcal{T}^\pi Z_1(\tilde{s}, \tilde{a}), \mathcal{T}^\pi Z_2(\tilde{s}, \tilde{a})) \\ &\leq \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma \sup_{\tilde{s}'_{i+j}, \tilde{a}'_{i+j}} d_p(Z_1(\tilde{s}'_{i+j}, \tilde{a}'_{i+j}), Z_2(\tilde{s}'_{i+j}, \tilde{a}'_{i+j})) \\ &= \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma \bar{d}_p(Z_1, Z_2) \end{aligned} \quad (28)$$

The weights p_i^o and p_j^a are normalized: $\sum_{i=1}^{\delta_a} p_i^a = 1$, $\sum_{j=1}^{\delta_o} p_j^o = 1$. Combining these weights with the discount factor, we define an effective discount factor:

$$\gamma_{\text{eff}} = \sum_{i=1}^{\delta_a} p_i^a \sum_{j=1}^{\delta_o} p_j^o \gamma \in (0, 1). \quad (29)$$

For any Z_1, Z_2 , the operator satisfies:

$$\bar{d}_p(\mathcal{T}^\pi Z_1, \mathcal{T}^\pi Z_2) \leq \gamma_{\text{eff}} \cdot \bar{d}_p(Z_1, Z_2), \quad (30)$$

By Banach's fixed-point theorem, $\mathcal{T}^\pi$ has a unique fixed point Z^π , and the sequence $Z_i = \mathcal{T}^\pi Z_{i-1}$ will converge to it as $i \rightarrow \infty$, i.e., Z_i will converge to Z^π as $i \rightarrow \infty$.

Lemma 3: (Policy Improvement): Let π_{new} be the optimal solution of maximizing cumulative returns. Then, $Q^{\pi_{new}}(\tilde{s}, \tilde{a}) \geq Q^{\pi_{old}}(\tilde{s}, \tilde{a})$ for $\forall(\tilde{s}, \tilde{a}) \in \mathcal{S} \times \mathcal{A}$.

Proof. We can update the policy by maximizing the objective in terms of Q value

$$\pi_{new}(\cdot|\tilde{s}) = \arg \max_{\pi} \mathbb{E}_{\tilde{a} \sim \pi} [Q^{\pi_{old}}(\tilde{s}, \tilde{a})] \quad \forall \tilde{s} \in \mathcal{S} \quad (31)$$

and the expected stochastic delay representation operator $\mathcal{T}_\pi$ can be written as

$$\mathcal{T}^\pi Q^\pi(\tilde{s}, \tilde{a}) = \sum_{i=1}^{\delta_a} p_i^a \cdot \sum_{j=1}^{\delta_o} p_j^o \cdot \left[\mathbb{E}_{\tilde{r}_{i+j} \sim \tilde{R}_{i+j}} [\tilde{r}_{i+j}(\tilde{s}_{i+j})] + \gamma \mathbb{E}_{\substack{\tilde{s}_{i+j+1} \sim p \\ \tilde{a}_{i+j+1} \sim \pi}} [Q^\pi(\tilde{s}_{i+j+1}, \tilde{a}_{i+j+1})] \right]. \quad (32)$$

Then, we can obtain that

$$\mathbb{E}_{\tilde{a} \sim \pi_{new}} [Q^{\pi_{old}}(\tilde{s}, \tilde{a})] \geq \mathbb{E}_{\tilde{a} \sim \pi_{old}} [Q^{\pi_{old}}(\tilde{s}, \tilde{a})] \quad \forall \tilde{s} \in \mathcal{S}. \quad (33)$$

By utilizing Eq. (33) and repeatedly expanding $Q^{\pi_{old}}$ on the right side with Eq. (32), we can derive the following results:

$$\begin{aligned} Q^{\pi_{old}}(\tilde{s}, \tilde{a}) &= \sum_{i=1}^{\delta_a} p_i^a \cdot \sum_{j=1}^{\delta_o} p_j^o \cdot \left[\mathbb{E}_{\tilde{r}_{i+j} \sim \tilde{R}_{i+j}} [\tilde{r}_{i+j}(\tilde{s}_{i+j})] + \gamma \mathbb{E}_{\substack{\tilde{s}_{i+j+1} \sim p \\ \tilde{a}_{i+j+1} \sim \pi_{old}}} [Q^{\pi_{old}}(\tilde{s}_{i+j+1}, \tilde{a}_{i+j+1})] \right] \\ &\leq \sum_{i=1}^{\delta_a} p_i^a \cdot \sum_{j=1}^{\delta_o} p_j^o \cdot \left[\mathbb{E}_{\tilde{r}_{i+j} \sim \tilde{R}_{i+j}} [\tilde{r}_{i+j}(\tilde{s}_{i+j})] + \gamma \mathbb{E}_{\substack{\tilde{s}_{i+j+1} \sim p \\ \tilde{a}_{i+j+1} \sim \pi_{new}}} [Q^{\pi_{old}}(\tilde{s}_{i+j+1}, \tilde{a}_{i+j+1})] \right] \\ &\dots \\ &\leq Q^{\pi_{new}}(\tilde{s}, \tilde{a}) \quad \forall(\tilde{s}, \tilde{a}) \in \mathcal{S} \times \mathcal{A} \end{aligned} \quad (34)$$

Thus, through the policy improvement, we can obtain $Q^{\pi_{new}}(\tilde{s}, \tilde{a}) \geq Q^{\pi_{old}}(\tilde{s}, \tilde{a})$ for $\forall(\tilde{s}, \tilde{a}) \in \mathcal{S} \times \mathcal{A}$.

We define π_k as the policy at iteration k . From Lemma 2, we can obtain the Z^{π_k} for $\forall \pi_k$ through stochastic delay representation with distributional returns policy evaluation process. Since the expectation of distributional value function $Z(\tilde{s}, \tilde{a})$ is the value $Q(\tilde{s}, \tilde{a})$, i.e. $Q(\tilde{s}, \tilde{a}) = \mathbb{E}[Z(\tilde{s}, \tilde{a})]$, we can obtain $Q^{\pi_k}(\tilde{s}, \tilde{a}) = \mathbb{E}[Z^{\pi_k}(\tilde{s}, \tilde{a})]$. By Lemma 3, $Q^{\pi_k}(\tilde{s}, \tilde{a})$ is monotonically increasing for $\forall(\tilde{s}, \tilde{a}) \in \mathcal{S} \times \mathcal{A}$. Since Q^π is bounded everywhere for $\forall \pi$, the policy sequence π_k converges to some $\pi^\dagger$ as $k \rightarrow \infty$, and it follows that

$$\mathbb{E}_{\tilde{a} \sim \pi^\dagger} [Q^{\pi^\dagger}(\tilde{s}, \tilde{a})] \geq \mathbb{E}_{\tilde{a} \sim \pi} [Q^{\pi^\dagger}(\tilde{s}, \tilde{a})] \quad \forall \pi \quad \forall \tilde{s} \in \mathcal{S}. \quad (35)$$

Utilizing the results of Lemma 3, we can obtain

$$Q^{\pi^\dagger}(\tilde{s}, \tilde{a}) \geq Q^\pi(\tilde{s}, \tilde{a}) \quad \forall \pi \quad \forall(\tilde{s}, \tilde{a}) \in \mathcal{S} \times \mathcal{A}. \quad (36)$$

Therefore, $\pi^\dagger$ is optimal, i.e., $\pi^\dagger = \pi^*$.

It is worth mentioning that Eq. (4) is applicable to environments where both observation and action delays are present and can be simplified to Eq. (5) when only one type of delay is present in the environment. Thus, Eq. (5) exhibits the same convergence.

C EXPERIMENTAL SETUP

C.1 Random Delay Environments

To better evaluate the algorithm’s performance, we design three delay distributions to simulate random delays in real application scenarios, as shown in Figure 6. The gamma delay distribution has a range of 1 to 6 with an expectation of 2, and the double Gaussian delay distribution has a range of 1 to 10 with an expectation of 5, while the uniform delay distribution has a range of 1 to 13 with an expectation of 6.

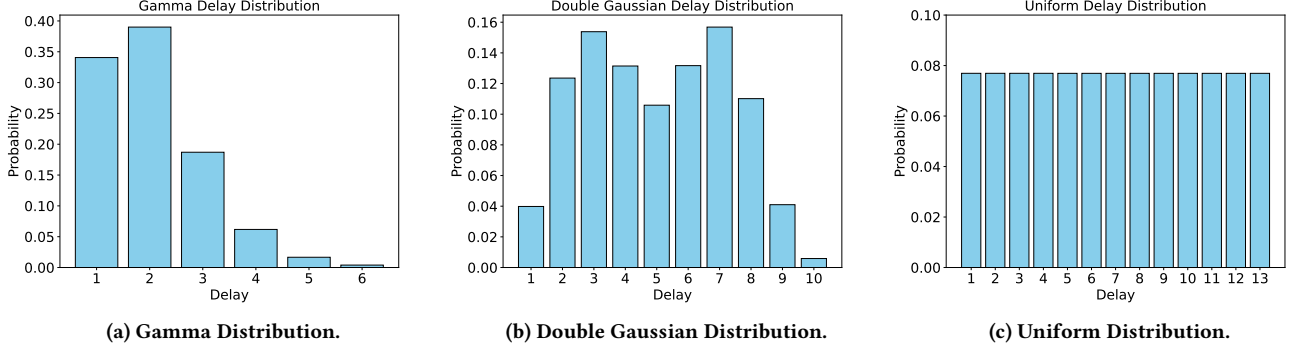


Figure 6: The different distributions of random delays.

C.2 Hyperparameters

Table 5: Hyperparameter Settings

Hyperparameter	Setting
Network	[256, 256, 256]
Batch Size	256
Total Timesteps	1,000,000
Learning Rate	0.0001
Learning Rate for α	0.0003
Hidden Activation	GELU
Output Activation	Linear
γ	0.99
Optimizer	Adam
Initial α	0.2

C.3 Compute Resources

All experimental results across our experiments are obtained on servers equipped with Intel(R) Xeon(R) CPU E5-2678 v3 @ 2.50GHz and NVIDIA GeForce RTX3090.

D EXPERIMENTAL RESULTS

D.1 Comparative Experimental Results in Random Delay Environments

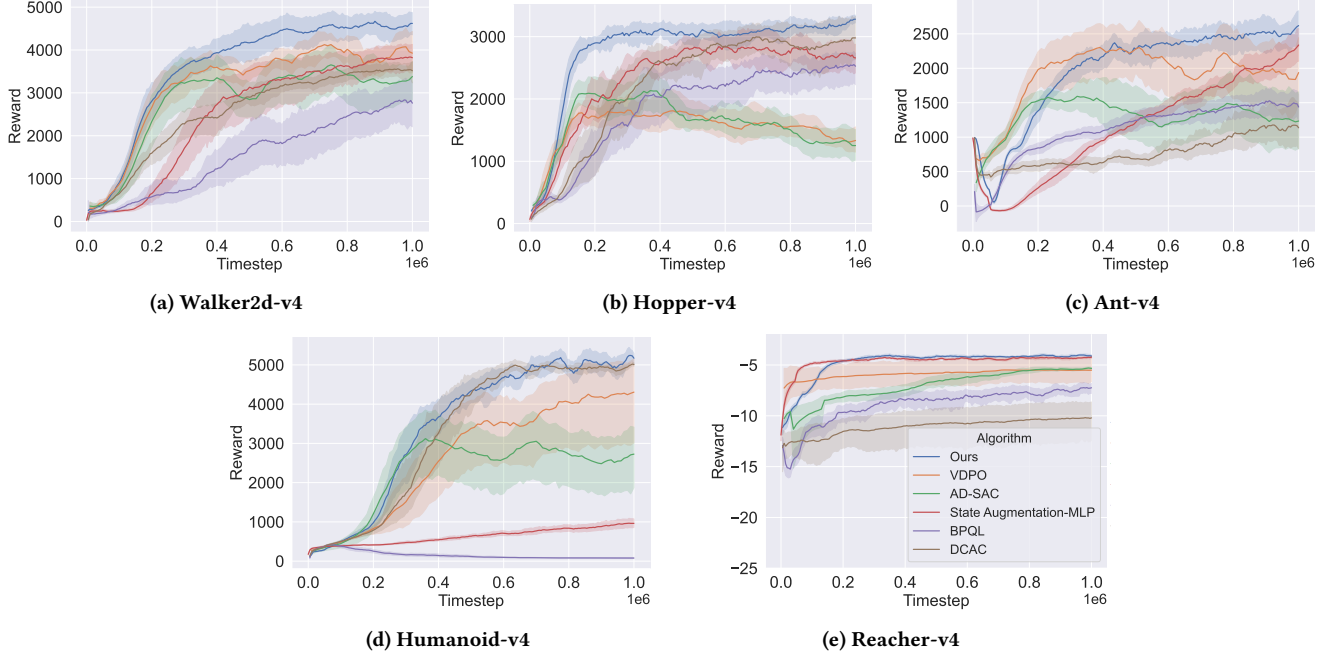


Figure 7: Comparison results in the gamma delay environment.

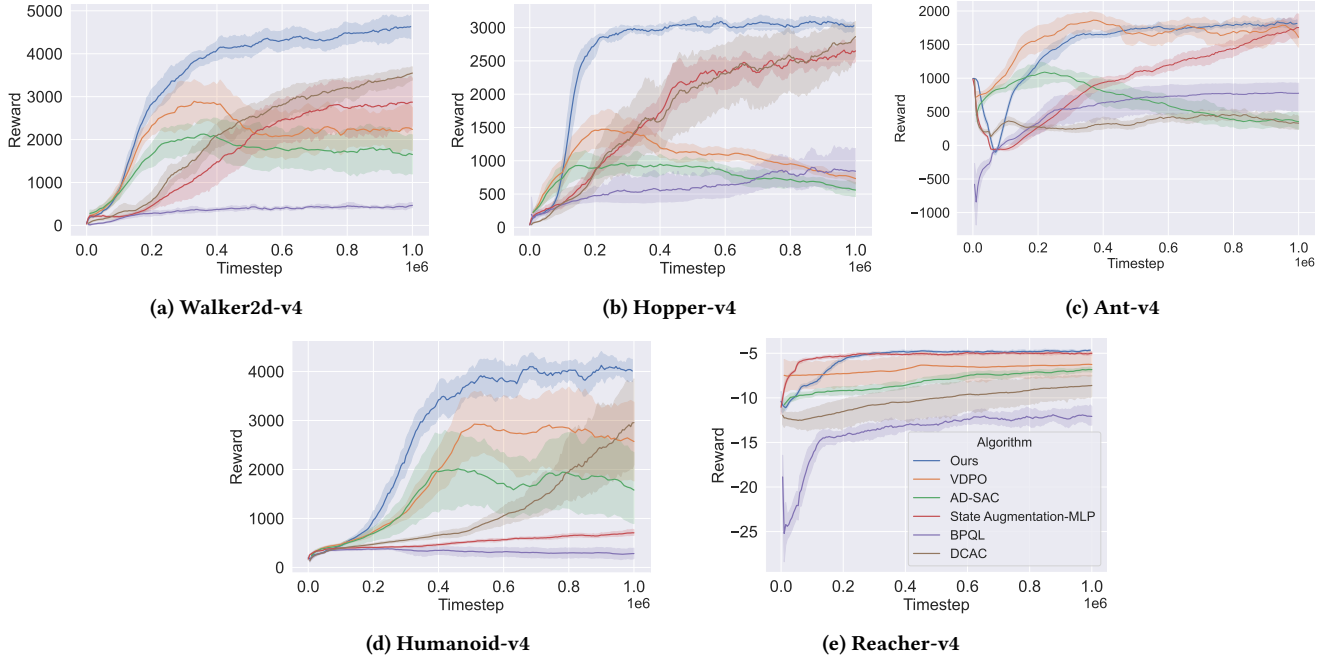


Figure 8: Comparison results in the double Gaussian delay environment.

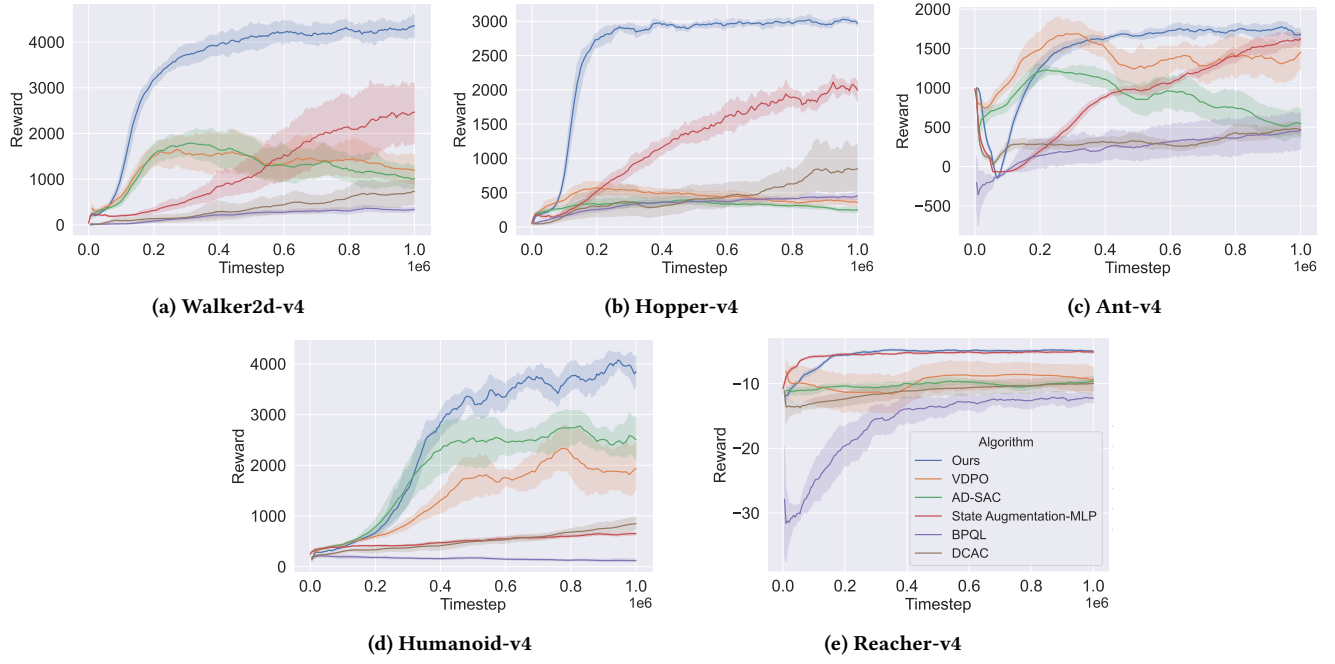


Figure 9: Comparison results in the uniform delay environment.

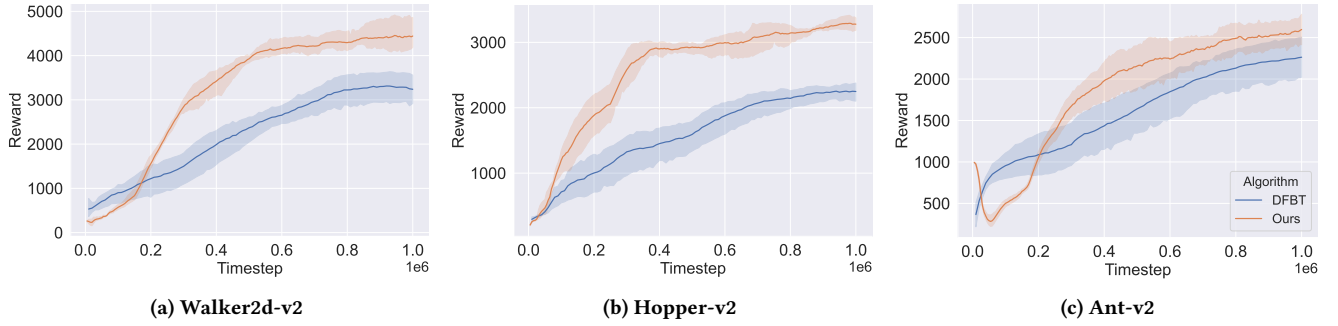


Figure 10: Additional Comparison results with DFBT in the gamma delay environment.

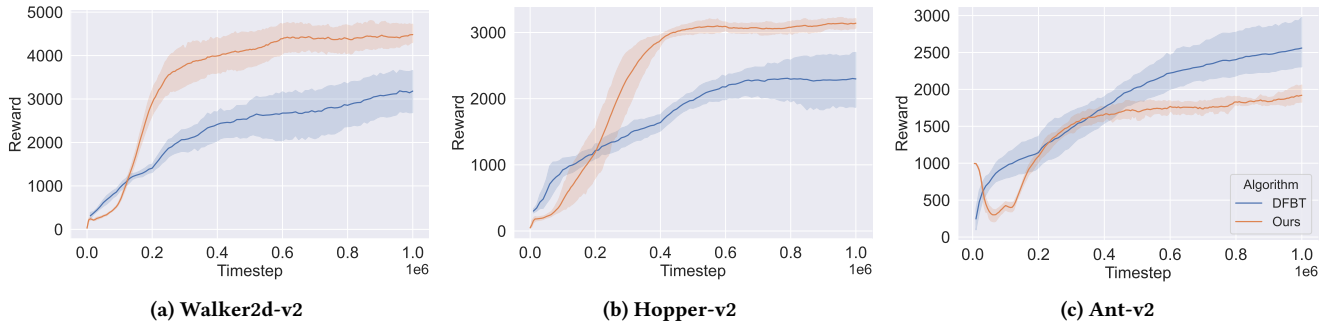


Figure 11: Additional Comparison results with DFBT in the double Gaussian delay environment.

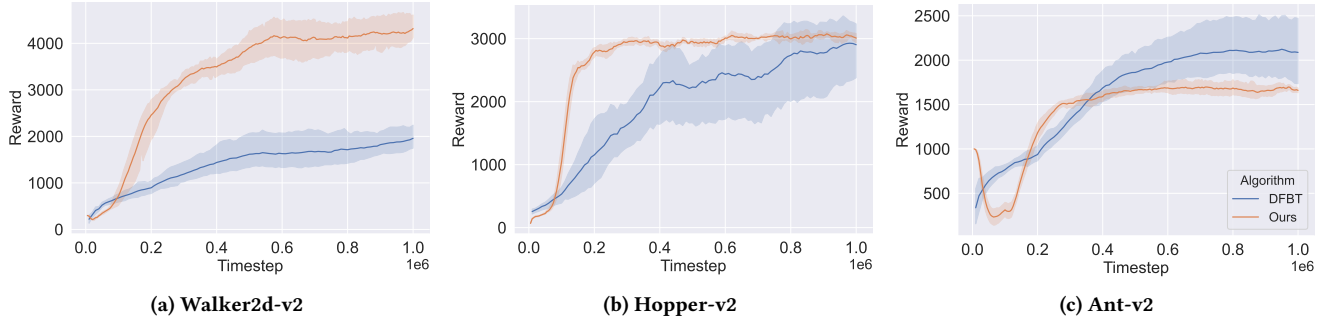


Figure 12: Additional Comparison results with DFBT in the uniform delay environment.

D.2 Comparative Experimental Results in a Constant Delay Environment

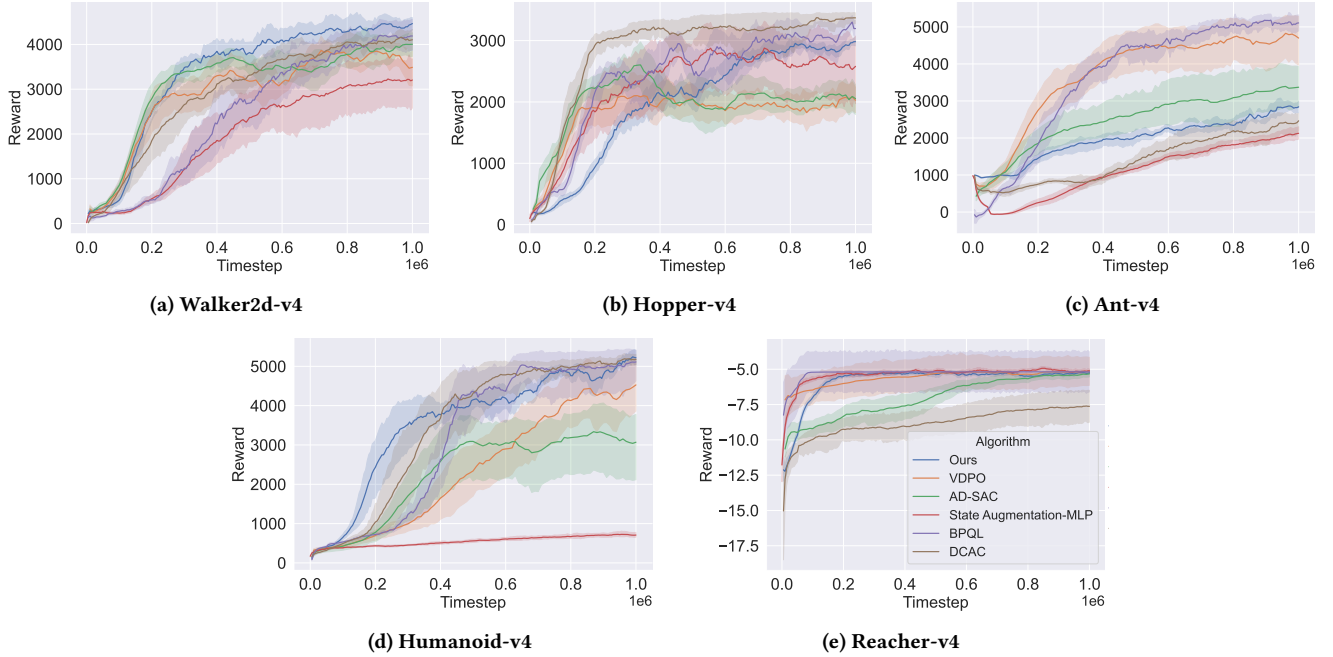


Figure 13: Comparison results in the constant-5 delay environment.

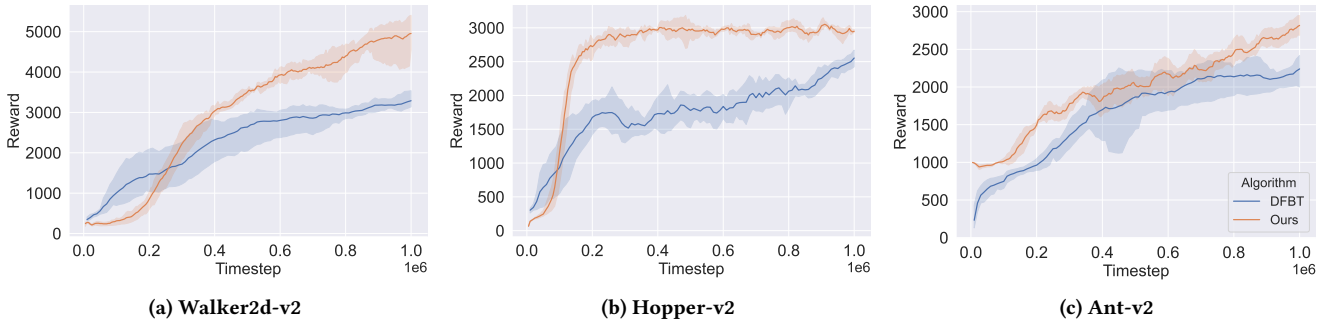


Figure 14: Additional Comparison results with DFBT in the constant-5 delay environment.

D.3 Evaluation Results of Handling Observation and Action Delays

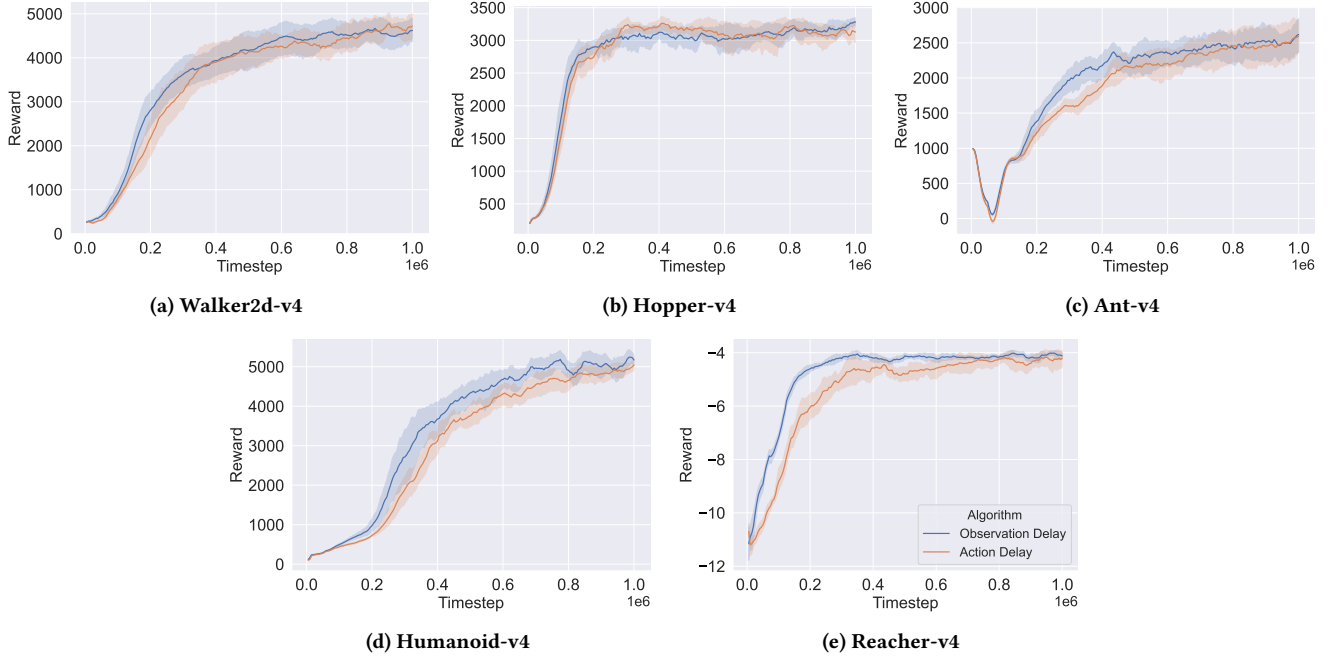


Figure 15: Evaluation Results of Handling Observation and Action Delays in gamma delay environment.

D.4 Ablation Experimental Results

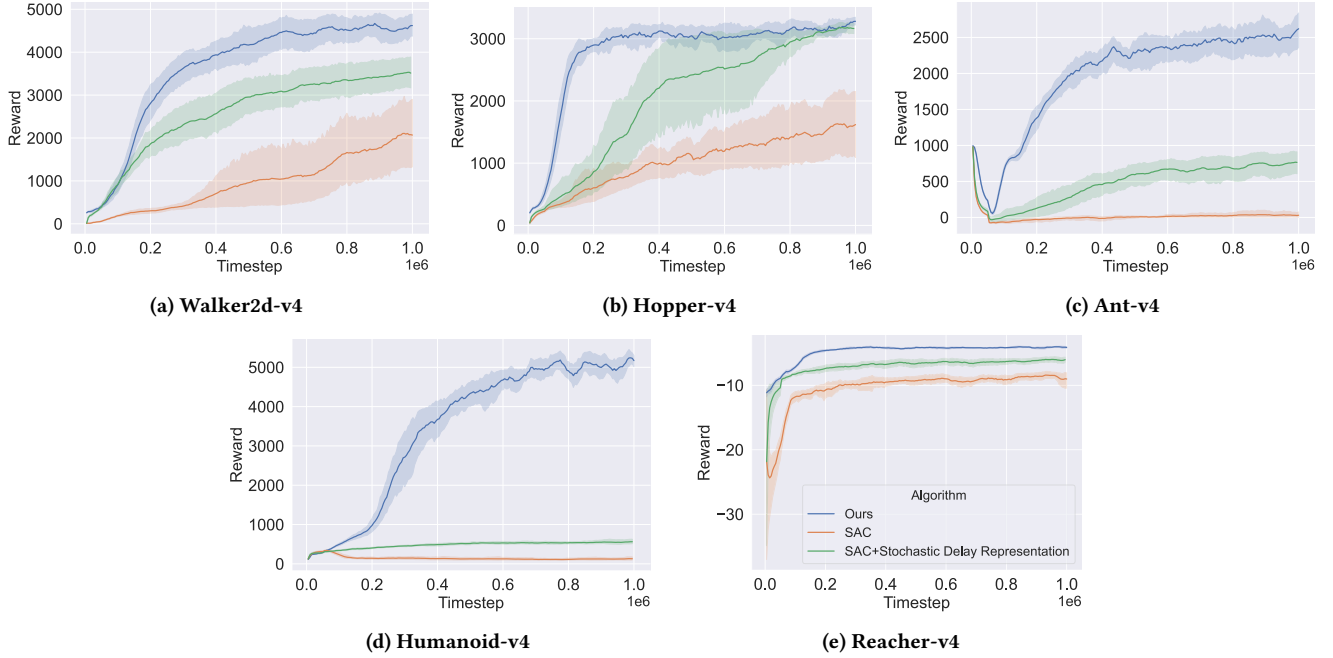


Figure 16: Ablation results in the gamma delay environment.

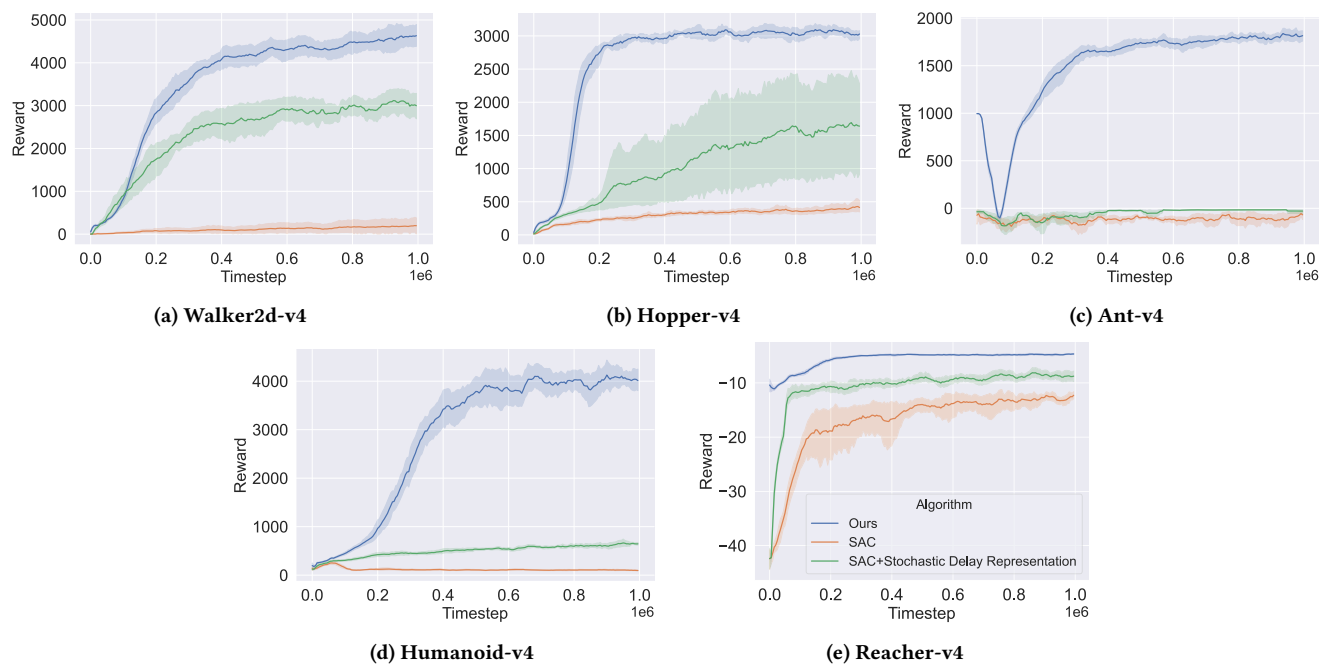


Figure 17: Ablation results in the double Gaussian delay environment.